CHAPTER 5

MARKET EQUILIBRIUM

CHAPTER ESSENCE AT A GLANCE:
Master market clearing price and quantity determination, simultaneous curve shifts, free entry zero-profit equilibrium, and government interventions (Price Ceiling and Price Floor).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Market Clearing Equilibrium:** Equilibrium occurs where market demand equals market supply ($Q_D = Q_S$); excess demand forces prices upward, while excess supply forces prices downward until equilibrium is restored.

• **Shifts in Demand and Supply Curves:** An increase in demand raises both equilibrium price and quantity; an increase in supply lowers equilibrium price and raises equilibrium quantity; simultaneous shifts depend on relative magnitudes.

• **Zero-Profit Long-Run Free Entry:** Under free entry and exit, firms enter if positive supernormal profits exist and exit if losses occur, driving market price down to minimum Long-Run Average Cost (P = min LAC).

• **Price Controls: Ceiling vs Floor:** A Price Ceiling sets a legal maximum price below equilibrium creating shortages and rationing; a Price Floor sets a legal minimum price above equilibrium creating surpluses requiring government procurement.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

$$\text{Market Equilibrium Condition} \iff Q_D(P^*) = Q_S(P^*) \rightarrow \text{Zero Excess Demand/Supply}$$
Market Clearing Price and Output Determination

$$\text{Price Ceiling} \iff P_c < P^* \rightarrow \text{Excess Demand (Shortage)} = Q_D - Q_S$$
Statutory Upper Price Limit Below Equilibrium

$$\text{Price Floor} \iff P_f > P^* \rightarrow \text{Excess Supply (Surplus)} = Q_S - Q_D$$
Statutory Minimum Price Above Equilibrium

$$\text{Free Entry Long-Run Price} \iff P^* = \min \text{LAC} \rightarrow \text{Zero Supernormal Profit}$$
Competitive Long-Run Sustainability Condition

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** A price ceiling is only binding and effective when set BELOW the equilibrium price; if set above equilibrium, market forces naturally prevail.

• **Alert:** A price floor (like Minimum Support Price) is only binding when set ABOVE the equilibrium price; it generates unsold surplus requiring state buffer stock purchases.

• **Alert:** Do not confuse excess demand with an outward shift of demand; excess demand is a disequilibrium condition at a given price where quantity demanded exceeds quantity supplied.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Market Equilibrium. Explain how excess demand and excess supply restore equilibrium. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Market equilibrium is a state of balance where quantity demanded by consumers equals quantity supplied by producers at the prevailing market price. **[1 Mark]**

• **Excess Demand:** Occurs when market price is below equilibrium price (QD > QS). Competition among buyers drives the price up, causing quantity demanded to contract and quantity supplied to expand until equilibrium is restored. **[1 Mark]**

• **Excess Supply:** Occurs when market price is above equilibrium price (QS > QD). Competition among sellers forces the price down, expanding quantity demanded and contracting quantity supplied until equality is reached. **[1 Mark]**

2. Explain the effects of a simultaneous rightward shift in both Demand and Supply curves on equilibrium price and quantity. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **General Effect on Quantity:** When both demand and supply increase simultaneously (rightward shifts), the equilibrium quantity increases unambiguously in all circumstances. **[1 Mark]**

• **Case 1 (Increase in Demand > Increase in Supply):** The upward pressure on price from demand exceeds downward pressure from supply; equilibrium price rises alongside quantity. **[1 Mark]**

• **Case 2 (Increase in Demand < Increase in Supply):** The downward pressure on price from abundant supply dominates; equilibrium price falls while quantity increases. **[1 Mark]**

• **Case 3 (Increase in Demand = Increase in Supply):** Upward and downward price pressures exactly cancel out; equilibrium price remains unchanged while quantity expands significantly. **[1 Mark]**

3. What is a Price Ceiling? Discuss its adverse consequences and remedies. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Definition:** A price ceiling is an imposed legal upper limit on the price of an essential commodity set by government strictly below equilibrium price to protect consumers (e.g. food grains, essential medicines). **[1.5 Marks]**

• **Adverse Consequences:**
• Creates chronic shortages as quantity demanded exceeds quantity supplied.
• Leads to black marketing, illegal hoarding, and deterioration of product quality. **[1.5 Marks]**

• **Policy Remedy:** Government introduces Dual Pricing and Rationing through the Public Distribution System (PDS) with ration cards to ensure equitable allocation of available quota. **[1 Mark]**

4. What is a Price Floor (Minimum Support Price)? Explain its economic impact. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** A price floor is a government-mandated legal minimum price set strictly above the equilibrium price to protect producer incomes (e.g. MSP for agricultural crops). **[1 Mark]**

• **Economic Impact:** Since price is above equilibrium, quantity supplied exceeds quantity demanded, creating a market surplus of unsold agricultural produce. **[1 Mark]**

• **Government Procurement:** To prevent distress selling, the government must purchase the entire surplus output at the support price and maintain buffer stocks for food security. **[1 Mark]**

5. Explain how equilibrium price is determined in a market with free entry and exit of firms. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Entry Mechanism:** If existing firms earn supernormal profits (P > AC), new firms enter the industry, increasing industry supply, shifting supply rightward, and lowering market price. **[1 Mark]**

• **Exit Mechanism:** If existing firms incur losses (P < AC), unprofitable firms exit, reducing market supply, shifting supply leftward, and raising market price. **[1 Mark]**

• **Long-Run Equilibrium:** Entry and exit cease only when market price equals minimum Long-Run Average Cost (P = min LAC), where each operating firm earns zero economic profit (normal profit). **[1 Mark]**

6. What happens to the equilibrium price of a commodity when its supply decreases while demand remains constant? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Direction of Change:** Equilibrium price rises while equilibrium quantity decreases. **[1 Mark]**

• **Adjustment Process:** A leftward shift of the supply curve creates excess demand at the original price; competition among buyers bids the price up to the new equilibrium. **[1 Mark]**